

Michael Matheny

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Summary

Senior Machine Learning Engineer / Applied Scientist with 10+ years of experience building large-scale AI and ML systems. **At Meta**, developed LLM evaluation and monitoring systems analyzing **10M+ daily responses** and **50K+ benchmark prompts**, improving dataset efficiency by 40% and reducing annotation costs by 30% through active learning. Previously **at Amazon**, built search ranking, recommendation, and fraud detection models that generated **\$400M+ revenue in 2022 and \$120M+ in 2023**, while reducing **\$8.3M in annual bad debt** and preventing **8 large-scale fraud attacks**. Holds a **Ph.D. in Computer Science from University of Utah**, with research focused on scalable algorithms and statistical modeling for large-scale data analysis.

Hard Skills

- **Programming Languages:** Python, SQL, JavaScript/TypeScript, Java, Scala, Bash
- **Machine Learning:** Supervised & Unsupervised Learning, Semi-Supervised Learning, Reinforcement Learning, Feature Engineering, Feature Selection, Model Evaluation, Hyperparameter Tuning, Ensemble Methods, Model Optimization
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, GRU, Transformers, Attention Mechanisms, Autoencoders, Transfer Learning
- **LLM & Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Fine-Tuning, Instruction Tuning, Prompt Engineering, RLHF, Embedding Models, Vector Database, LangChain, LangGraph, LLaMA, NLP, Agentic AI, Mlflow, Amazon Bedrock
- **Natural Language Processing:** BERT, GPT, Named Entity Recognition (NER), Text Classification, Sentiment Analysis, Question Answering, Chatbots, Tokenization, Speech-to-Text, Text-to-Speech
- **Computer Vision:** Image Classification, Object Detection, Image Segmentation, Feature Extraction, OpenCV, Graph WaveNet
- **Machine Learning Frameworks:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, XGBoost, LightGBM
- **Data Science & Analytics:** Pandas, NumPy, SciPy, Statistical Modeling, Hypothesis Testing, A/B Testing, Time Series Analysis, Bayesian Modeling, Data Cleaning, Data Wrangling
- **Data Engineering & Pipelines:** ETL Pipelines, Data Pipelines, Airflow, Feature Stores
- **Big Data & Distributed Systems:** Apache Spark, PySpark, Hadoop, Hive, Databricks, Amazon EMR, Snowflake
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, PostGIS, Amazon Redshift, NoSQL
- **Cloud & MLOps:** AWS (Bedrock, SageMaker, EC2, EKS, Lambda, S3, CloudWatch), GCP, Azure, Docker, Kubernetes, CI/CD, Jenkins, ML Pipelines, Model Deployment, Model Monitoring
- **Visualization:** Matplotlib, Seaborn, Plotly, Tableau

WORK EXPERIENCE

Meta | Seattle, WA

Senior ML Engineer | 03.2024 – Present

- Engineer on the **Meta AI Post-Training team**, developing large-scale systems to evaluate and improve the **factuality, safety, and reliability of Large Language Models** using **PyTorch**.
- Developed automated evaluation frameworks for LLMs with **image understanding and computer vision capabilities**, evaluating hallucinations, reasoning accuracy, factual correctness, NLP understanding, and caption quality across 50K+ prompts.

- Built **large-scale monitoring pipelines** processing **10M+ daily LLM responses**, automatically detecting and tracking failure modes such as hallucinations, toxicity, and reasoning errors.
- Developed an **active learning system** that prioritizes high-value samples for human annotation, improving **LLM post-training dataset efficiency by 40%** and reducing annotation costs by **30%**.
- Built scalable **LangChain-based LLM evaluation and FAISS-based vector retrieval pipelines** integrated with MLflow for model lifecycle management and evaluation tracking, reducing evaluation turnaround time by 60%.
- Partnered with research scientists and product teams to support **continuous LLM improvement cycles**, accelerating model iteration speed by **2×**.

Technologies: PyTorch, Transformers, LLM Evaluation, Active Learning, Distributed Data Pipelines, Spark, AWS, LangChain, LangGraph, MCP, Spark, Hugging Face, RAG, MLflow, Docker, Kubernetes, Computer Vision, A/B Testing, Redis

Amazon | Seattle, WA

Senior Applied Scientist | 06.2023 – 02.2024

- Developed **LLM-powered conversational product search and recommendation models** serving **50M+ monthly users**, improving product discovery relevance and increasing click-through rate by **7%** in controlled A/B experiments.
- Built **semantic retrieval pipelines leveraging Amazon Bedrock, transformer embeddings and vector databases** to rank over **500M+ products**, reducing query-to-product latency by **35%** while improving ranking accuracy by **12%**.
- Designed large-scale training and feature pipelines using **PySpark, Amazon EMR, and AWS SageMaker** to process **5TB+ daily behavioral and catalog data**, reducing model training time by **40%**.
- Led experimentation and causal analysis for GenAI shopping assistants, improving **conversion rate by 3.5%** across key e-commerce categories.

Technologies: PyTorch, Transformers, LLMs, PySpark, Amazon EMR, Recommendation Systems, Retrieval Systems, AWS, vector DB, Amazon Bedrock, AWS SageMaker

Applied Scientist II | 11.2020 – 06.2023

- Increased Amazon's gross revenue by **\$400M+ in 2022** through new optimized organic search models in the US and EU5 and **\$120M+ in 2023** through better autocomplete recommendations.
- Developed a **causal inference framework** to infer brand preferences, improving **personalization and recommendation relevance** in Amazon search.
- Led **fraud detection and chargeback automation** initiatives, reducing US bad debt by **\$8.3M annually** and saving **\$1.2M annually** in electronic gift card fraud.
- Detected and prevented **8 large-scale electronic gift card fraud attacks**, mitigating potential losses of **\$12K–\$6M per incident**.
- Mentored multiple scientists and **managed three ML interns**, delivering several end-to-end machine learning projects.

Technology: Deep Learning, Project Management, Transform, ETL Pipelines, Causal Inference, PySpark, Amazon Elastic MapReduce (EMR)

Applied Scientist I | 11.2019 – 11.2020

- Built **LightGBM and TensorFlow-based fraud detection models** for credit and gift card transactions, preventing **\$3M+ in losses** and reducing annual bad debt by **\$3.2M**.
- Designed and implemented **ETL pipelines processing ~2TB of daily transaction data**, enabling faster feature engineering and model iteration by **30%**.
- Leveraged **Amazon Redshift and PySpark** to aggregate and analyze millions of transactions per day, improving early detection of anomalous activity.

Technologies: Extract, Transform, Load (ETL), Amazon Redshift, PySpark, LightGBM, AWS SageMaker, Credit Card Fraud Prevention, Amazon Elastic MapReduce (EMR), Artificial Intelligence, Machine Learning, TensorFlow

University of Utah | Salt Lake City, UT

Research Assistant | 09.2015 – 10.2019

- Built an **AI-assisted traffic decision support system** for evacuation scenarios combining historical traffic data with near real-time roadway monitoring for congestion and incident awareness.
- Evaluated **spatio-temporal forecasting models**, including early graph-based approaches, achieving up to 23% improvement over **ARIMA** for short-term traffic prediction.
- Developed **attention-based sequence models for multi-horizon congestion forecasting**, improving early bottleneck detection and **reducing prediction error by 18%**.
- Built scalable traffic data pipelines using **Kafka, PySpark, and Snowflake** to process and analyze **500K+ daily GPS and roadway sensor records**, with Grafana dashboards for traffic analytics, monitoring, and congestion forecasting.
- Built **reinforcement learning models** for traffic signal control and evacuation routing, achieving **91% precision and 87% recall** for crash detection and improving evacuation throughput by 18% in SUMO simulations.

Technologies: Python, PyTorch, TensorFlow, scikit-learn, graph neural networks, sequence models, reinforcement learning, Kafka, Spark, AWS, PostGIS, Grafana, Snowflake

Sandia National Laboratories | Albuquerque, New Mexico Area

Graduate Research Intern | 05.2015 - 08.2015

- Developed and implemented **named entity recognition (NER) pipelines** using **spaCy** and **Stanford SENNA**, improving information extraction accuracy for research tasks.
- Processed and analyzed datasets using **NumPy and Pandas**, enabling scalable and reproducible data workflows.

Technologies: spaCy, NumPy, Pandas, NER, Stanford SENNA, NLP

University of Delaware | Newark, DE

Research Assistant | 06.2011 - 03.2014

- Developed an optimization framework for **choosing parameters for the ADIOS I/O library** on systems during heavy filesystem contention.
- Wrote software integrating learning games into the pre-existing volunteer computing project, **Docking@Home**.

Tutor | February 2011 - May 2012

- **Mentored** students in various computer science and math classes.

Oak Ridge National Laboratory | Oak Ridge, TN

Undergraduate Researcher | 06.2013 – 08.2013

- Worked on the **QMCPack software under Scott Klasky** where I developed better checkpointing methods and migrated the code to ADIOS.

Technologies: Python, SQL, PyMC3, Matplotlib, Hierarchical Modeling, Hadoop

Education

University of Utah | Salt Lake City, CA - Ph.D, Computer Science 2014 – 2019

University of Delaware | Berkeley, CA - Bachelor's Degree, Computer Science 2010 – 2014